



ETHICS AND ECONOMICS OF INSTITUTIONAL GOVERNANCE
Winter Term 2022/23, 1st Trial

- Please answer **all** questions. Maximum: **60 points**, time to answer the questions: **60 minutes**
- In the case of cheating or deception, the exam will be graded as failed.
- Please answer in complete sentences, not bullet points. Indicate the problem number with your answers.

Problem 1: Lecture 10 & 11 (literature, 30 points)

Both Bjorn Lomborg and William Nordhaus focus on climate change in their research. Please, describe their main arguments by answering the following sub-questions:

- a) Criticizing the mainstream approach to fight global warming, Bjorn Lomborg draws public attention to two alternative strategies how to deal with the challenges of climate change. Please explain.
(5 points)
- b) Bjorn Lomborg argues that society gets climate change wrong because of media bias. At the same time, he does not think that journalists conspire to lead the democratic public astray. Please explain. Make use of an economic model to analyze the social dilemma between a representative journalist and the rest of media.
(10 points)
- c) William Nordhaus criticizes the political practice of setting an arbitrary temperature target for global climate policy. What is his main argument? (5 points)
- d) According to William Nordhaus, the core problem of addressing climate change can be described as a global public good. What are the main implications of such a description? (5 points)
- e) Why does William Nordhaus propose to mitigate the climate change problem with the help of a universal carbon price that should increase in the course of time? Please explain (5 points)



Problem 2: European Union Emissions Trading System (lecture 11, 30 points)

Despite having brought to existence a European emissions trading system (EU ETS), European governments (e.g., Germany) still adopt additional measures to reduce emissions, e.g. by promoting green energy. Please reconstruct the effects of such measures on emissions trading from an economic point of view by answering the following questions. Please, draw suitable diagrams of the market for emissions in order to support your answers.

- a) Please, characterize the initial EU ETS market equilibrium without any additional measures by an individual government (e.g., Germany). Please, draw an appropriate market diagram and verbally explain the initial situation as an interplay of supply and demand. (6 points)
- b) Now, the German government implements additional measures (e.g., EEG - renewable energies act) with the aim to reduce the domestic demand for CO₂ emissions. Please explain with the help of an appropriate diagram the implications for the collective demand curve and effects on quantity and price. (9 points)
- c) What is the climate impact of the German efforts to promote green energy (via EEG)? Please distinguish between emissions on German territory and emissions on European territory. (5 points)
- d) Please, explain why and how this problem is duplicating on a global scale with regard to a very inelastic worldwide supply of fossil fuels. Please, draw an appropriate market diagram and explain the nature of this global “green paradox” phenomenon. What are the implications for a purely demand-oriented climate policy? (10 points)